

A ADDITIONAL PROOFS

THEOREM 4.2. The query selection problem (Problem 2), i.e., finding the optimal subset $S^* = \arg \max_{S \subseteq Q, |S| \geq k} \frac{h(S, \tau)}{|S|}$, is NP-hard.

PROOF. Step 1: Properties of $h(S, \tau)$. Recall that $h(S, \tau)$ counts the number of dissimilar pairs (q_i, q_j) in S such that $\text{sim}(q_i, q_j) < \tau$. We note the following:

- $h(S, \tau)$ is *monotone*: if $A \subseteq B$, then $h(A, \tau) \leq h(B, \tau)$, because adding more elements cannot decrease the number of dissimilar pairs.
- $h(S, \tau)$ is *submodular*: the marginal gain of adding a new query q to a smaller set A is at least as large as adding it to a larger set $B \supseteq A$, since some dissimilar pairs involving q may already be counted in B .

Thus, $h(\cdot, \tau)$ is a monotone submodular function.

Step 2: Relation to submodular optimization. Our objective is to maximize

$$\frac{h(S, \tau)}{|S|}, \quad \text{subject to } |S| \geq k.$$

This is equivalent to maximizing $h(S, \tau)$ under a normalization factor $|S|$. Since $|S| \geq k$, the denominator does not vanish. Therefore, the problem belongs to the family of *submodular-over-cardinality* optimization problems.

Step 3: Reduction from Minimum Submodular Cover. The *Minimum Submodular Cover* problem is defined as follows: given a monotone submodular function $f : 2^Q \rightarrow \mathbb{Z}_{\geq 0}$, an integer k , and a threshold α , decide whether there exists a set $S \subseteq Q$ with $|S| \geq k$ such that $f(S) \leq \alpha$. This problem is known to be NP-hard.

We now reduce this problem to our query selection problem. Consider an arbitrary instance (f, Q, k, α) of Minimum Submodular Cover. Construct the corresponding instance of our problem by defining:

$$h(S, \tau) := C - f(S),$$

where C is a sufficiently large constant (e.g., $C = \max_{S \subseteq Q} f(S)$).

Since f is monotone submodular, so is h . Maximizing $\frac{h(S, \tau)}{|S|}$ with $|S| \geq k$ is then equivalent to minimizing $f(S)$ with $|S| \geq k$, up to the additive constant C and scaling by $|S|$.

In particular, if we could solve our problem in polynomial time, we could also solve Minimum Submodular Cover in polynomial time.

Step 4: Conclusion. Since Minimum Submodular Cover is NP-hard, and our problem generalizes it through the above reduction, it follows that computing the exact optimal solution S^* to the query selection problem is NP-hard.

Thus, unless $P=NP$, no polynomial-time algorithm exists for solving this problem optimally. $\square$

THEOREM 4.3. Given a query set Q , a budget k , and a threshold τ , the algorithm Greedy returns a set S with $|S| \geq k$ and $\frac{h(S, \tau)}{|S|} \geq \frac{1}{2} \times \frac{h(S^*, \tau)}{|S^*|}$, where S^* is an optimal solution to the Problem 2.

PROOF. Let S^* be an optimal solution of size at least k , i.e.,

$$S^* \in \arg \max_{|S| \geq k} \frac{h(S, \tau)}{|S|}.$$

Algorithm 1 constructs a sequence $S_0 = \emptyset$, $S_j = S_{j-1} \cup H_j$, where

$$H_j \in \arg \max_{T \subseteq Q \setminus S_{j-1}} \frac{h(S_{j-1} \cup T, \tau) - h(S_{j-1}, \tau)}{|T|},$$

and stops at the first index ℓ with $|S_\ell| \geq k$. Here we define

$$g_j(X) := h(S_j \cup X, \tau) - h(S_j, \tau).$$

We distinguish two cases here.

Case 1. There exists $j \in \{1, \dots, \ell\}$ such that

$$g_{j-1}(S_j \cap S^*) \geq \frac{1}{2} h(S^*, \tau).$$

By monotonicity,

$$h(S_j, \tau) \geq h(S_{j-1}, \tau) + g_{j-1}(S_j \cap S^*).$$

After padding S_j to size at least k ,

$$\frac{h(S'_j, \tau)}{|S'_j|} \geq \frac{h(S_j, \tau)}{k} \geq \frac{g_{j-1}(S_j \cap S^*)}{k} \geq \frac{h(S^*, \tau)}{2k}.$$

Since $k \leq |S^*|$, we obtain

$$\frac{h(S, \tau)}{|S|} \geq \frac{1}{2} \times \frac{h(S^*, \tau)}{|S^*|}.$$

Case 2. For every $j \in \{1, \dots, \ell\}$ we have

$$g_{j-1}(S_j \cap S^*) < \frac{1}{2} h(S^*, \tau).$$

This means that at each step, at least half of the value $h(S^*, \tau)$ remains outside S_{j-1} . By the choice of H_j , we then have

$$\rho_{S_{j-1}}(H_j) \geq \frac{1}{2} \frac{h(S^*, \tau)}{|S^*|}.$$

Hence,

$$h(S_j, \tau) - h(S_{j-1}, \tau) \geq \frac{1}{2} \frac{h(S^*, \tau)}{|S^*|} \times |H_j|.$$

Summing from $j = 1$ to $\ell - 1$ gives

$$h(S_{\ell-1}, \tau) \geq \frac{1}{2} \frac{h(S^*, \tau)}{|S^*|} \times |S_{\ell-1}|.$$

By monotonicity, $h(S_\ell, \tau) \geq h(S_{\ell-1}, \tau)$ and $|S_\ell| \geq k$. Thus,

$$\frac{h(S_\ell, \tau)}{|S_\ell|} \geq \frac{h(S_{\ell-1}, \tau)}{|S_\ell|} \geq \frac{1}{2} \frac{h(S^*, \tau)}{|S^*|} \times \frac{|S_{\ell-1}|}{|S_\ell|} \geq \frac{1}{2} \frac{h(S^*, \tau)}{|S^*|}.$$

Finally, the algorithm compares all candidates S'_j (padded to size k) and returns the one with the highest diversity ratio. Therefore, the algorithm achieves a $\frac{1}{2}$ -approximation ratio. $\square$

B ADDITIONAL DISCUSSIONS

B.1 Prompts

In this subsection, we provide the key prompts used in our work.

Table 9: Parameter sets used in our experiments.

Module	Parameter	Options	Naive RAG	Advanced RAG
Chunking	Chunk size	128, 256... 1024	✓	
	Split method	recursive, sentence, token	✓	
	Overlap ratio	0.0, 0.05, 0.1, ..., 0.5	✓	
Embedding	Embedding model	bge-m3 [4] bge-large-en-v1.5 [22] Finance2_embedding_small-en-V1.5 [2] bge-base-en-v1.5 [22] bge-small-en-v1.5 [22] granite-embedding-125m-english [2] multilingual-e5-large [19] ¹ gte-large [13] thenlper-gte-base [13]	✓	
Query decomposing	Subquery num	2, 3, 4, 5		✓
	Fusion mode	simple reciprocal_rerank dist_based_score relative_score		✓
Retrieving	Retrieval method	dense, sparse, hybrid	✓	✓
	Num chunks	1, 2, 3, ..., 10	✓	
	Hybrid weight	0.1, 0.2, 0.3, ..., 0.9		✓
Reranking	Reranker model	flashrank [6] TransformerRanker [9] MonoT5 [16] RankT5 [26] MonoBERT [15] InRanker [12] EchoRank [1]		✓
		Top k		✓
		Content enhance		✓
Generating	Prompt template	default, concise, CoT	✓	✓

Prompt for extracting high-level insights**Insight Extraction Instruction:**

You are an expert RAG system analyst. Your task is to analyze a specific RAG configuration along with its representative query executions, then produce insights that explain its performance patterns and give actionable recommendations.

Analysis Requirements:

1. Identifies Key Success/Failure Factors.
2. Explains Component Interactions.
3. Provides Actionable Recommendations.
4. Highlights Query Type Patterns.

Input Placeholder:

Dataset information: [Info]
Current configuration: [Configuration]
Representative queries: [Query1][Query2][...]

B.2 Limitations

In our work, we devoted significant effort to designing a practical and reliable RAG tuning system. However, our study still has several limitations, primarily due to resource constraints.

(1) Limited Knowledge Datasets. Our evaluation involved eight representative datasets, whereas RAG systems are widely applied across many other domains. Expanding the dataset coverage could provide a more comprehensive understanding of system generalization. (2) Resource Constraints. Due to computational limits, the largest model we used was Qwen2.5-72B, and the overall experiments consumed nearly 10 billion tokens. Running even larger models—such as GPT-5 (175B parameters) or Gemini Ultra—would dramatically increase the cost, potentially reaching hundreds of thousands of dollars. While more powerful models may further improve performance, the 72B model already offers a strong balance between capability and feasibility. Exploring larger models remains an important direction for future work. (3) Prompt Sensitivity. The

Figure 14: The prompt for extracting high-level insights.

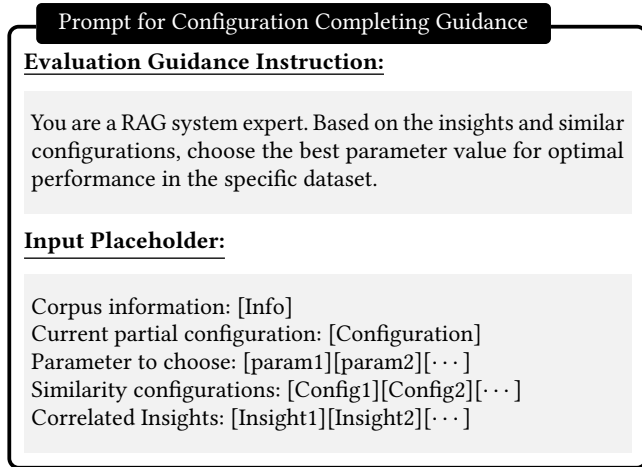


Figure 15: The prompt for configuration completing guidance.

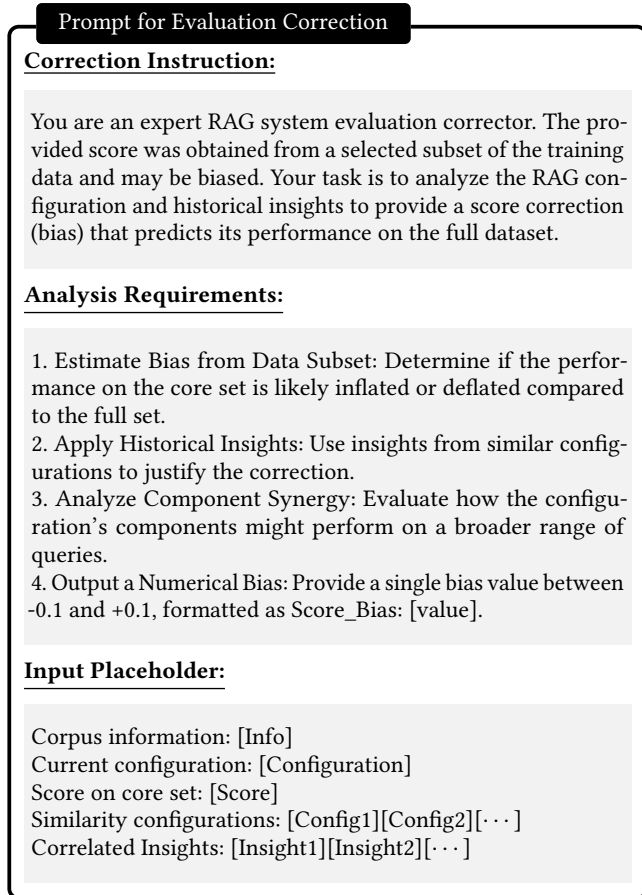


Figure 16: The prompt for evaluation correction.

performance of all methods is highly sensitive to prompt design. Because of resource limitations, we did not conduct prompt ablation

studies and instead adopted prompt templates from prior works. (4) RAG Architecture Scope. Our study focuses on the standard modular RAG pipeline. Other variants, such as graph-based RAGs [25], are not included. Nonetheless, we believe our key ideas, particularly the memory-guided tuning framework, can be readily adapted to those architectures with minimal modification.

These limitations highlight promising directions for future exploration. Addressing them would enable a more comprehensive and reliable evaluation of RAG tuning methodologies and further advance research in this area.

B.3 How to employ coreset in RAG tuning

Indeed, our query selection problem can be viewed as an instance of the *coreset selection* problem. Both aim to select a representative subset S from a large dataset Q (where $|S| \ll |Q|$) to approximate the performance on the full set. That is, the existing coreset selection algorithm can also be applied here. In this context, given a query set Q and a size constraint k , the coreset selection problem is defined as follows:

$$S = \arg \min_{S \subseteq Q} \sum_{i=1}^{|Q|} \min_{q_j \in S} \|\mathbf{v}_i - \gamma_j \cdot \mathbf{v}_j\|, \quad \text{s.t. } |S| \leq k, \quad (11)$$

where $\mathbf{v}_i$ denotes the embedding vector of query q_i , and γ_j denotes the number of queries in Q that are represented by the selected query $q_j \in S$. Specifically, a query q_i is considered to be represented by q_j if their semantic similarity exceeds a predefined threshold.

B.4 Future works

There are many promising directions for future work on RAG systems and RAG tuning.

First, an interesting avenue is to explore multi-modal retrieval-augmented generation, extending current text-based pipelines to incorporate visual, tabular, and code modalities for richer contextual grounding.

Second, integrating agentic RAG frameworks into the traditional RAG pipeline could enable adaptive reasoning, tool use, and decision-making, allowing the system to dynamically plan retrieval and generation strategies.

Third, it would be valuable to study the transferability of RAG tuning across models, domains, and tasks, enabling efficient adaptation and reducing the cost of re-tuning for new scenarios.

C ADDITIONAL EXPERIMENTS

C.1 Dataset

Specifically, the datasets include:

- **MedQA** [11]: originated from the medical domain and contains highly specialized questions, designed to test a model's accuracy in a specific knowledge area.
- **FiQA** [23]: a conversational QA dataset in the financial domain, requiring the model to not only understand financial reports but also handle conversational context.
- **HotpotQA** [24]: a multi-hop reasoning dataset based on Wikipedia, where each question requires integrating information from multiple paragraphs to infer the correct answer.
- **2WikiMultiHopQA** [10]: another multi-hop reasoning dataset that extends the concept of HotpotQA by explicitly linking entities across two different Wikipedia articles, enabling more complex reasoning chains.

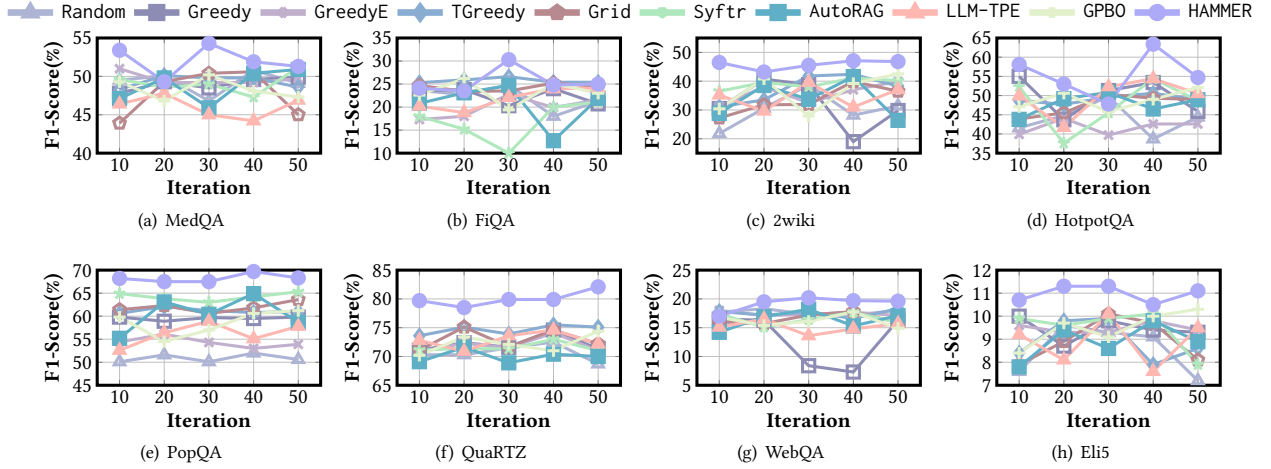


Figure 17: F1-Score variance with iterations.

Table 10: Performance comparison of HAMMER variants.

Model	FiQA			2Wiki			HotpotQA					QuaRTZ	
	F1	Faithfulness	ROUGE-L	F1	EM	Accuracy	F1	EM	Accuracy	Faithfulness	ROUGE-L	F1	Accuracy
HAMMER-Mem	25.1	55.6	12.0	39.9	30.6	42.3	51.2	40.0	60.4	82.7	56.1	72.5	84.4
HAMMER-Sim	21.5	58.2	12.5	42.3	31.9	46.6	56.0	39.2	58.4	84.3	50.3	77.8	83.5
HAMMER-Eva	27.4	58.5	9.5	41.6	30.5	45.2	52.1	40.2	57.0	80.5	50.4	75.4	81.5
HAMMER	30.3	62.7	13.6	47.1	37.7	50.6	63.4	44.4	66.3	86.9	63.3	82.1	86.3

Table 11: Performance comparison of HAMMER variants.

Model	FiQA			2Wiki			HotpotQA					QuaRTZ	
	F1	Faithfulness	ROUGE-L	F1	EM	Accuracy	F1	EM	Accuracy	Faithfulness	ROUGE-L	F1	Accuracy
Qwen2.5-7B	30.3	62.7	13.6	47.1	37.7	50.6	63.4	44.4	66.3	86.9	63.3	82.1	86.3
DeepSeek-R1-32B	29.6	63.2	15.2	47.3	37.5	54.5	65.1	48.4	65.5	89.3	60.0	88.0	94.1
Qwen-2.5-72B	28.9	58.5	10.1	52.9	44.1	62.9	68.6	50.6	73.2	91.6	65.5	89.5	94.9
GPT-4o-mini	31.4	73.3	18.2	53.7	42.6	58.5	67.1	49.7	69.8	89.1	65.9	88.5	94.9

- **PopQA** [14]: focuses on open-domain questions about popular entities, designed to evaluate factual recall and general world knowledge.
- **Quartz** [18]: contains questions involving qualitative reasoning in science, requiring commonsense and conceptual understanding.
- **Web Questions** [3]: a classic dataset where answers are sourced from general web pages, testing a model’s open-domain question answering ability.
- **ELI5** [7]: a long-form QA dataset that requires the model to generate detailed, easy-to-understand explanations for complex questions sourced from social media.

C.2 The configuration space

To systematically explore the optimal configurations for each module in the RAG system, we construct a comprehensive parameter search space covering all components of the pipeline, as summarized in Table 9.

- **1) Chunking.** We explore three splitting methods [21]: (i) recursive character splitting, which attempts to keep semantically related text contiguous; (ii) sentence-based splitting, which respects grammatical boundaries; and (iii) token-based splitting, which produces fixed-size chunks. Concurrently, we tune the chunk size across four values and the overlap ratio across eleven values, as these parameters directly influence the granularity and contextual continuity of retrieved content.
- **2) Embedding.** To convert text into vector representations for semantic retrieval, we evaluate nine mainstream embedding models, such as bge-m3 [4] and gte-large [13], to assess how different semantic representation capabilities affect downstream performance.
- **3) Retrieving.** We test three retrieval methods: (i) dense retrieval, which identifies semantically similar content; (ii) sparse retrieval using keyword matching methods like BM25 [17]; and (iii) hybrid retrieval, which combines both to leverage semantic similarity and keyword relevance. We also vary the number of retrieved documents across ten options to balance recall and computational efficiency.

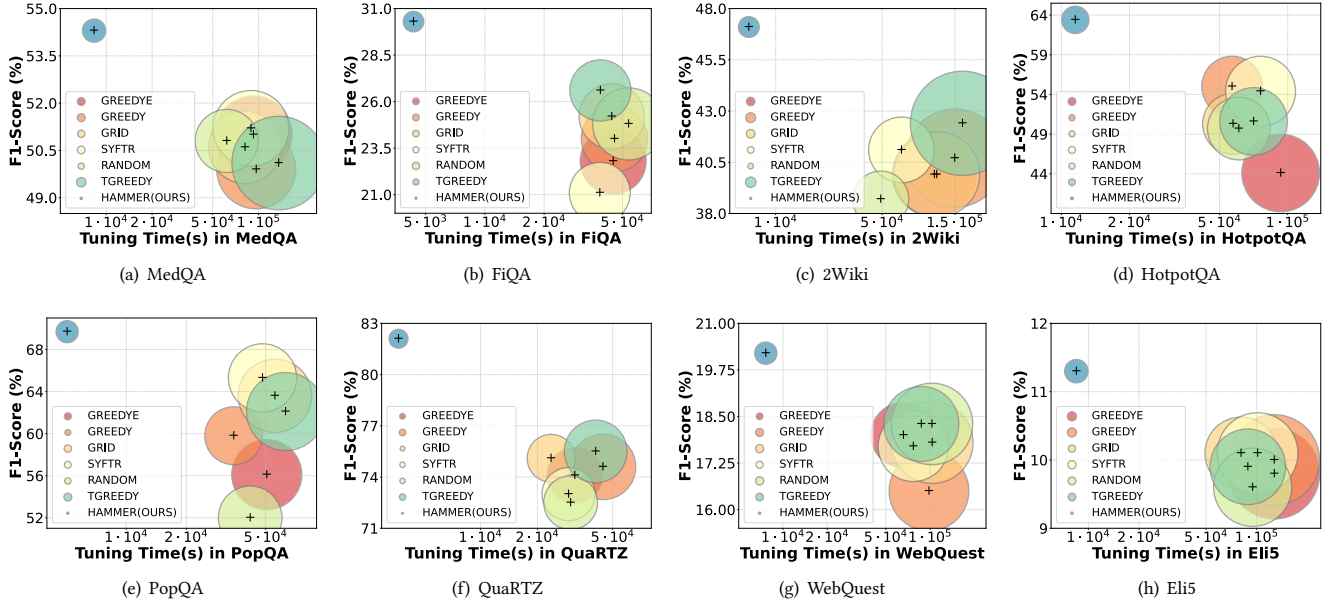


Figure 18: Tuning Time vs. F1-Score across datasets.

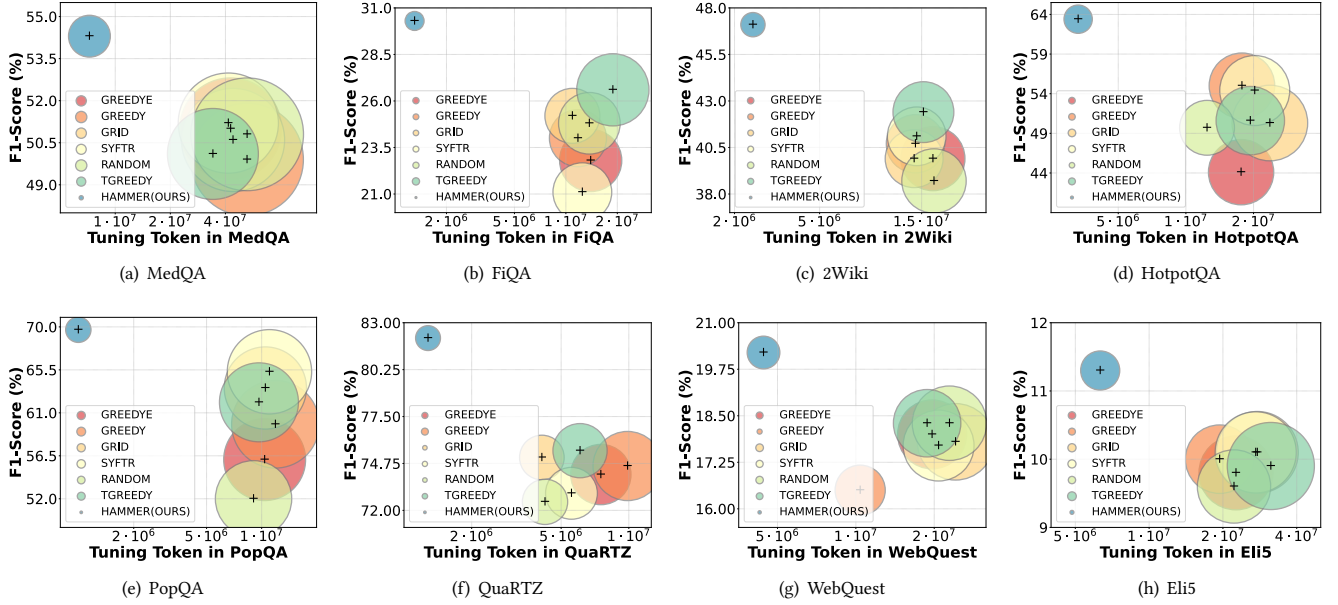


Figure 19: Tuning token vs. F1-Score across datasets.

- 4) *Query decomposing*. To handle complex multi-hop queries, we employ query decomposition [8] to divide them into simpler sub-questions. We further explore four fusion strategies for combining sub-results, including Reciprocal Rank Fusion [5].
- 5) *Reranking*. Following retrieval, we apply seven state-of-the-art reranker models [26] to re-score and reorder candidate documents. We also tune the number of retained documents across 31 possible values and explore context enhancement by including adjacent chunks to enrich retrieved information.

- 6) *Generating*. For the response generation stage, we utilize three distinct prompt templates to structure model inputs: (i) a *default mode* for direct answering, (ii) a *concise mode* for brevity, and (iii) a *Chain-of-Thought (CoT) mode* [20] to encourage step-by-step reasoning.

C.3 Additional results

In this subsection, we provide the additional results.

► **Exp.1. Ablation study.** In this experiment, we conduct an ablation study for HAMMER by progressively removing its key memory-guided components. Specifically, we create two variants: HAMMER-Sim, which removes the memory-guided simulation component, and

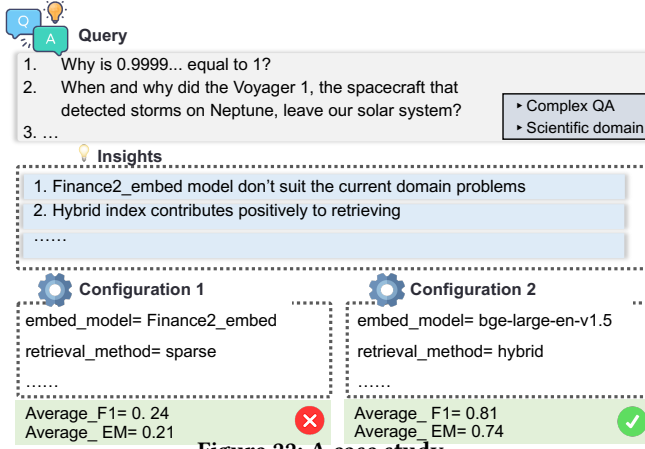


Figure 22: A case study.

Table 12: Analysis of query selection strategies.

Method	HotpotQA (F1)			2Wiki (F1)		
	Random	Syfttr	Greedy	TGreedy	Syfttr	Greedy
Full	49.7	54.4	55.0	42.4	41.1	40.7
Select	47.3	50.2	50.4	40.1	38.9	39.0
Select+Random	45.1	47.7	46.9	38.2	37.3	37.9

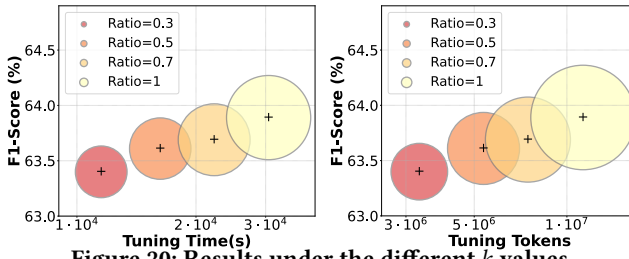
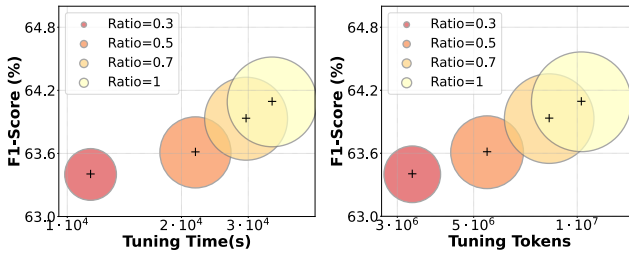
Figure 20: Results under the different k values.

Figure 21: Results under the different sample query sizes.

HAMMER-Eva, which removes the memory-guided evaluation component. The results, shown in Table 10, reveal that: 1) Both HAMMER-Sim and HAMMER-Eva perform worse than the full HAMMER, confirming that both components contribute effectively. The former replaces the random rollout with memory-guided simulation, enabling more informed exploration, while the latter refines the evaluation process using insights from SE-Bank, ensuring more reliable feedback. 2) Notably, HAMMER-Sim, which still evaluates configurations using the entire query set, performs worse than HAMMER. This indicates that our memory-guided evaluation not only approximates the full evaluation effectively but also refines it through LLM-based correction, achieving even higher reliability and efficiency.

► **Exp.2. Overall comparison of tuning efficiency and effectiveness.** We evaluate different RAG tuning methods in terms of

tuning time, token consumption, and the corresponding F1-scores across all datasets, as shown in Figures 18 and 19. Our method, HAMMER, consistently achieves the best overall performance—requiring the least tuning time and tokens while obtaining the highest F1-scores. This superiority stems from our carefully designed query selection mechanism and the hierarchical three-layer graph memory structure, SE-Bank, which jointly enable efficient and knowledge-driven RAG tuning.

► **Exp.3. Effect of Hyper-parameters.** In HAMMER, two hyperparameters are critical: the selection size k and the size of the random query subset B_Q . By default, both k and $|B_Q|$ are set to 30% of the available query set. To examine their influence, we vary these parameters and report the results in Figures 20 and 21. In these figures, ratio=0.x indicates that the parameter k or $|B_Q|$ is set to $0.x \times |Q|$. The observations are as follows: 1) As the sizes of k and B_Q increase, the performance of HAMMER also improves, which is intuitive — using more queries during evaluation provides a more accurate estimation of configuration quality. 2) However, the improvement remains marginal even when using the full query set compared to only 30% of it (e.g., F1-score increases from 63.5 to 64.5), while the token and time costs increase by about fourfold. This demonstrates that the proposed query selection and memory-guided evaluation are highly effective — even a small subset of representative queries, when combined with knowledge-guided refinement, is sufficient to achieve near-optimal tuning performance with much lower cost.

► **Exp.4. Case study.** We present a case study to show that the distilled insights play a key role in finding the optimal configuration. As shown in Figure 22, we compare two configurations on the ScienceQA dataset. The insights reveal that the Finance2_embed model is not a suitable choice for this dataset, while the hybrid retrieval strategy demonstrates higher effectiveness. We observe that Configuration 2, which follows these insights, achieves significantly better performance than Configuration 1, which does not.

► **Exp.5. Analysis of query selection.** The reduction in token cost and tuning time in HAMMER stems from two complementary designs: 1) before tuning begins, HAMMER employs a query selection module to identify a representative subset of queries; 2) during each tuning iteration, HAMMER adopts a memory-guided evaluation mechanism that evaluates only a small batch of queries and refines the resulting feedback using knowledge stored in SE-Bank.

We further isolate the effect of query selection by constructing three variants for each method: Full, which performs tuning on the complete query set; Select, which integrates our query selection module; and Select+Random, which applies query selection but replaces memory-guided evaluation with random partial evaluation in each iteration. As shown in Table 12, using query selection alone leads to only a moderate performance drop compared to full-query tuning, indicating that the selected queries remain representative. In contrast, Select+Random suffers a substantial F1-score degradation despite achieving significant cost reductions. These results demonstrate that achieving high-performance configurations at low tuning cost is not driven by query selection alone, but critically relies on the proposed memory-guided mechanisms and the structured knowledge stored in SE-Bank.

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